

PAPER387 — Relativistic SCm Velocity Parameter Update: $v_{SCm} = 0.99c$

Session: 106 (groksharecfdcad2f5.txt full analysis)

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Source: groksharecfdcad2f5.txt, lines ~9600–10122 (main.cpp global constants) **Section:** Star Magic_construction file_040ct2025.docx — global parameter declarations **Session:** 106 (groksharecfdcad2f5.txt full analysis) **CP4 Class:** vSCmRelativisticParameterUpdateCalculator (CP4 #38)

1. Overview

Prior UQFF implementations used a preliminary value $v_{SCm} = 1 \times 10^8$ m/s for the Superconductive medium velocity parameter in the reactive energy term E_{react} . The Star Magic_construction file_040ct2025.docx Grok thread formalizes an updated value:

$$v_{SCm} = 0.99 \times c = 0.99 \times 2.998 \times 10^8 \text{ m/s} = 2.968 \times 10^8 \text{ m/s}$$

This represents the first formal assignment of v_{SCm} to a relativistic speed grounded in observational evidence from the J1610+1811 quasar jet ($z=3.122$, covered in PAPER_374).

PAPER374 identified $v=0.99c$ as the jet velocity for J1610+1811. PAPER375 used this in the UQFF wormhole/Meissner advanced integration context. However, **neither paper formally updated the v_{SCm} constant in the core parameter set** — this paper makes that explicit and calculates the cascading impact on the E_{react} term.

2. The v_{SCm} Parameter

2.1 Definition

v_{SCm} is the characteristic velocity of the Superconductive medium (SCm) phase in UQFF. It appears in the reactive energy term connecting quantum vacuum density to the SCm-plasma coupling:

$$E_{\text{react}} = \frac{\rho_{SCm}}{e^{\{-\kappa t\}}} \cdot v_{SCm}^2 \cdot \rho_A$$

Where:

ρ_{SCm} = SCm vacuum density (kg/m^3)

v_{SCm} = SCm characteristic velocity (m/s)

ρ_A = ambient density (kg/m^3), default $\rho_A = 1 \times 10^{23} \text{ kg/m}^3$

κ = decay constant, default $\kappa = 0.0005 \text{ day}^{-1}$

t = time elapsed

2.2 Parameter Update: Old vs New

Parameter	Old Value	New Value	Ratio
v_{SCm}	$1 \times 10^8 \text{ m/s}$	$2.968 \times 10^8 \text{ m/s (0.99c)}$	$2.968\times$
v_{SCm}^2	$1 \times 10^{16} \text{ m}^2/\text{s}^2$	$8.808 \times 10^{16} \text{ m}^2/\text{s}^2$	$8.808\times$

The velocity-squared amplification factor is **8.808×**, meaning all `Ereact` calculations using the prior value are underestimated by approximately one order of magnitude.

3. Observational Basis

The update is validated by the J1610+1811 relativistic quasar jet:

- System:** J1610+1811, $z=3.122$ (lookback time ~ 11.5 Gyr)
- Jet power:** $P_{\text{jet}} \approx 4 \times 10^{45} \text{ W}$
- Jet velocity:** $v = 0.99c$
- Lorentz factor:** $\gamma = (1 - v^2/c^2)^{-1/2} \approx 7.089$
- Source documents:** Star Magic_09Sept2025.docx, Star Magic_construction file_04Oct2025.docx

This system demonstrates that SCm-driven plasma jets reach 0.99c, making this the physically motivated upper-bound velocity for the SCm velocity parameter.

4. Updated Ereact Calculation

For the canonical SGR1745 parameters:

$\rho_{\text{SCm}} \approx \rho_A = 1 \times 10^{-23} \text{ kg/m}^3$
 $t = 0$ (initial condition)

Old:

$E_{\text{react}}^{\text{old}} = \frac{(1 \times 10^{-23})(1 \times 10^{16})}{1 \times 10^{-23}} \cdot e^0 = 1 \times 10^{16} \text{ J/m}^3$

New:

$E_{\text{react}}^{\text{new}} = \frac{(1 \times 10^{-23})(8.808 \times 10^{16})}{1 \times 10^{-23}} \cdot e^0 = 8.808 \times 10^{16} \text{ J/m}^3$

The reactive energy increases by a factor of **8.808×** across all systems.

5. Global Constants Context (Oct 2025)

The full confirmed global parameter set from `main.cpp` (Oct 2025 build):

```
const double c = 2.998e8; // Speed of light (m/s)
double v_SCm = 0.99 * c; // SCm velocity = 2.968e8 m/s ← UPDATED
double Omega_g = 7.3e-16; // Galactic angular velocity (rad/s)
double Mbh = 8.15e36; // SMBH mass (kg)
double dg = 2.55e20; // Galaxy scale distance (m)
double rho_A = 1e-23; // Ambient density (kg/m³)
double rho_sw = 8e-21; // Solar wind density (kg/m³)
double v_sw = 5e5; // Solar wind velocity (m/s)
```

```
double kappa = 0.0005; // UQFF decay constant (day-1)
double alpha = 0.001; // Coupling constant α
double gamma = 0.00005; // Coupling constant γ
double k1=1.5, k2=1.2, k3=1.8, k4=2.0; // MUGE layer weights
double beta_i = 0.603; // Buoyancy coupling (≈0.6 calibrated)
double rho_v = 6e-27; // Vacuum energy density (kg/m³)
double C_concentration = 1.0; // Concentration factor
double f_feedback = 0.1; // Feedback fraction
const double num_strings = 1e9; // String count
```

6. Implications for UQFF Pipeline

6.1 Affected Equations

- Ereact term:** All systems using v_{SCm^2} scaling see 8.808× amplification
- Compressed MUGE:** The v_{SCm^2}/c^2 relativistic correction factor changes from 0.1111 (old) to 0.9801 (new) — approaching unity
- Lorentz correction:** With $v=0.99c$, the Lorentz factor $\gamma=7.089$ is now accessible for relativistic corrections in jet-class systems

6.2 Calibration Compatibility

The calibrated constant $\kappa=0.0005/\text{day}$ (from PAPER_341) remains unchanged — the decay envelope is independent of the velocity amplitude.

The calibrated $\beta_i\approx0.603$ (PAPER_375) also remains valid as it governs buoyancy coupling, not the SCm velocity channel.

7. Validation Cross-Reference

Reference	Connection
PAPER_374	J1610+1811 jet physics providing the $v=0.99c$ basis
PAPER_375	UQFF Wormhole/Meissner integration using $\gamma=7.089$
PAPER_341	$\kappa=0.0005/\text{day}$ calibration (unchanged by this update)
PAPER_372	Compressed MUGE 8-term base (Ereact channel)

8. Canonical Value (All Future Implementations)

$v_{SCm} = 0.99c = 2.968\times10^8 \text{ m/s}$ is the canonical Superconductive medium velocity.

All UQFF Python and C++ implementations should use:

```
c = 2.998e8 # m/s
v_SCm = 0.99 * c # = 2.968e8 m/s

const double c = 2.998e8;
double v_SCm = 0.99 * c; // = 2.968e8 m/s
```

Discovery Class: Parameter Formalization — First explicit canonical assignment of $v_{\text{SCM}}=0.99c$ **Distinct from:** PAPER374 (*J1610 jet observational context*); PAPER375 (Meissner/wormhole use of γ) **Impact:** 8.808× amplification of all Ereact-channel UQFF calculations